

## Episode 44: One More Time About *Hakaba Kitaro* – Image Resolution Talk

I somehow managed to get through the turbulent tax return process, and finally have some free time. This winter I was constantly busy with *Hakaba Kitaro*, so I kept putting off hot pot promises with friends and drinking party invitations, but now I'm going to do lots of that!

Though with it getting this warm, hot pot might be questionable? Borderline? (bitter laugh)

By the way, yesterday I went to a Genghis Khan restaurant that's supposedly delicious. And man, it was even more delicious than expected! The lamb had that famous Hokkaido restaurant taste—so good! I ate like crazy and went home in a great mood.

But you know, just like with yakiniku, Genghis Khan makes your clothes and hair smell. Even more than yakiniku because it's lamb. So today, before leaving, I thoroughly scrubbed myself all over in the shower, and of course changed all my clothes before coming to the studio. But wait? I could still faintly smell something delicious...

Ah! I messed up! (sweat)

The upper half was perfectly taken care of, but actually, I wore the same jeans (sweat). When I sat down in my chair at my desk, from around my thighs, there was a faint...

Hmm, I let my guard down! Genghis Khan is formidable. But I'll go eat it again!

Well then.

*Hakaba Kitaro*, currently airing to great acclaim—last week's episode 10 apparently got a 5.8% rating in the Kanto region. Wow, 5.8%! And the peak instantaneous rating was

7.3%. Since I usually work on productions that don't really have anything to do with ratings, I get a little excited (laugh).

By the way, to put that amazing number in perspective—it's about as impressive as ratings that Kimutaku's dramas get (laugh).

Despite such a late time slot, we're truly happy that so many people are watching. As I've written before, what makes us production staff happiest is having people actually watch our work. Thank you all so much.

Production of *Hakaba Kitaro* is now completely finished for all episodes, with only DVD jacket work remaining.

So today, let's talk about screen image resolution.

Not just *Hakaba Kitaro*, but Toei Animation productions are basically made at 144dpi resolution. The basic key animation size (standard animation paper size) fits on A4 paper, and in terms of data size, for *Hakaba Kitaro*, the basic colouring data is "1596 pixels × 898 pixels," made a bit larger than the photography frame. Backgrounds are the same. They're also digitised at 144dpi. We combine these, apply filters, add textures, vary line weights, and do various other things in photography to create those screens.

144dpi seems to be a high setting industry-wise. Toei has used this resolution since digital colouring began, but other production companies' works vary quite a bit—120, 105, 95, and so on. Higher resolution naturally means larger data size, which affects work efficiency depending on the computer specs used for work. Especially for photography staff—the more screen effect specifications there are, the incredibly long it takes to shoot (composite) one cut.

But well, recently basic machine specs have improved dramatically, and with more productions being made in HD for terrestrial digital compatibility, 144dpi seems to be becoming the industry mainstream.

This 144dpi is sufficient resolution for standard frame cuts, but for camera work cuts that zoom way in on small frames, even 144dpi makes the image rough. In such cases, we need to work at even higher resolutions, but that makes colouring tough. When data gets too large, colouring software doesn't run smoothly. Photography staff face the same difficulties.

Sometimes directors and assistants give instructions for simple calculations based on frame reduction ratios—288 or 300. One or two frames might be manageable, but when 30 or 40 consecutive frames are made such high resolution, it basically becomes impossible to work with. Actually, you can get sufficiently beautiful screens without such high resolution (bitter laugh).

However, the story is different for posters and still images. There's a different way of thinking between data for moving screens and static images for print materials.

Unlike footage with movement, posters and jackets are static and can be examined closely by hand, so we finish them at high resolution and take care that lines don't show jaggedness.

Also, this type of copyright-related imagery can be used in unpredictable ways. For example, if it's truly limited to DVD jacket material, something just a bit larger than the main story—288dpi or so—would be totally fine, but sometimes it suddenly gets blown up large and used for promotional posters in stores, leading to "Oh no! (crying)" situations. So I make this type of copyright material on the large side (bitter laugh).

The first key visual for *Hakaba Kitaro*—that poster of baby Kitaro emerging from the earth—was created with screen work at 400dpi using A3-sized paper for the key

animation. The two stills for "Shokotan's" CD jacket were also each done with B4-sized paper key animation, also at 400dpi. And the DVD jacket is likewise B4 size at 400dpi. All were coloured, combined with backgrounds, and given the same photography processing as the main story images. Incidentally, all are original drawings by character designer Yamamuro-san.

Compared to main story cut data, these are incredibly large image data files, so when applying photography processing, the texture sizes were completely inadequate, making things quite difficult apparently.

Finally, you can see DVD volume 1 jacket images on the web and such. That flat, old print-like image is actually the outer jacket design, while the jacket for the soft vinyl package itself is made with that same artwork but using the exact colouring and photography processing from the main story. It's turned out quite suspicious (mysterious). Please look forward to it.

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